Report for Take Home 2:

G06
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Checklist:

- ☑ ADD
- ☑ SUB
- ☑ AND
- ☑ OR
- ☑ XOR
- ☑ NOT
- ☑ SHL
- ☑ SHR
- ☑ SLT
- ☑ Tri-State Buffer

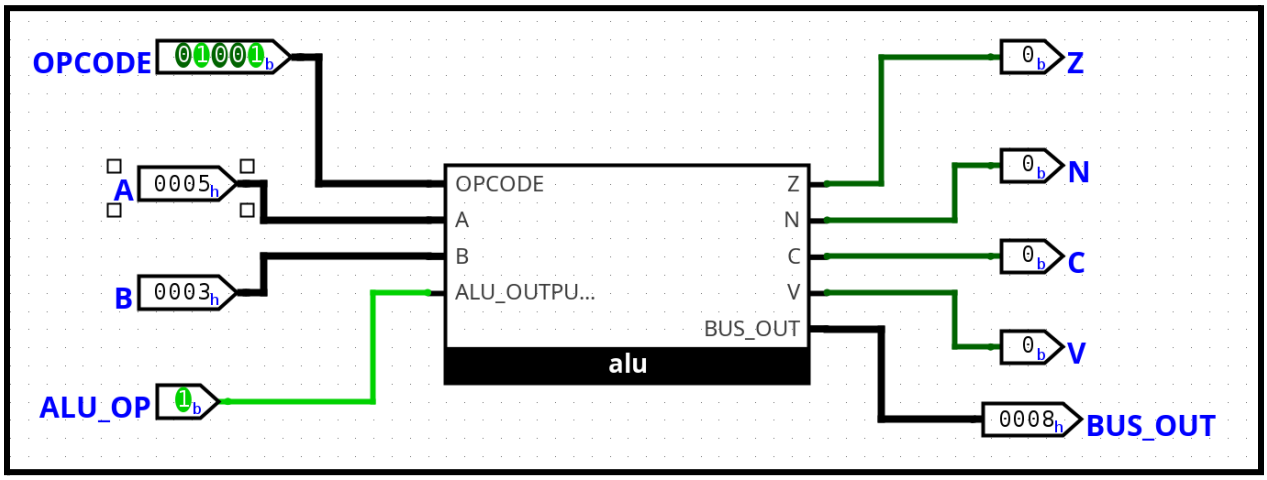
Interface:

Signal	Direction	Width	Usage
`A, B`	Input	16 bits each	2-inputs to the ALU
`OPCODE`	Input	5 bits	5-bit code for determining which operation to perform
`ALU_OUTPUT_ENABLE`	Input	1 bit	1 if ALU is required for computation
`BUS_OUT`	Output via tri-state buffer	16 bits	Gives the output if the ALU is Enabled
`ZERO, NEGATIVE, CARRY, OVERFLOW`	Output	1 bit each	Gives 1 if value is 0, < 0, has carry and overflow respectively

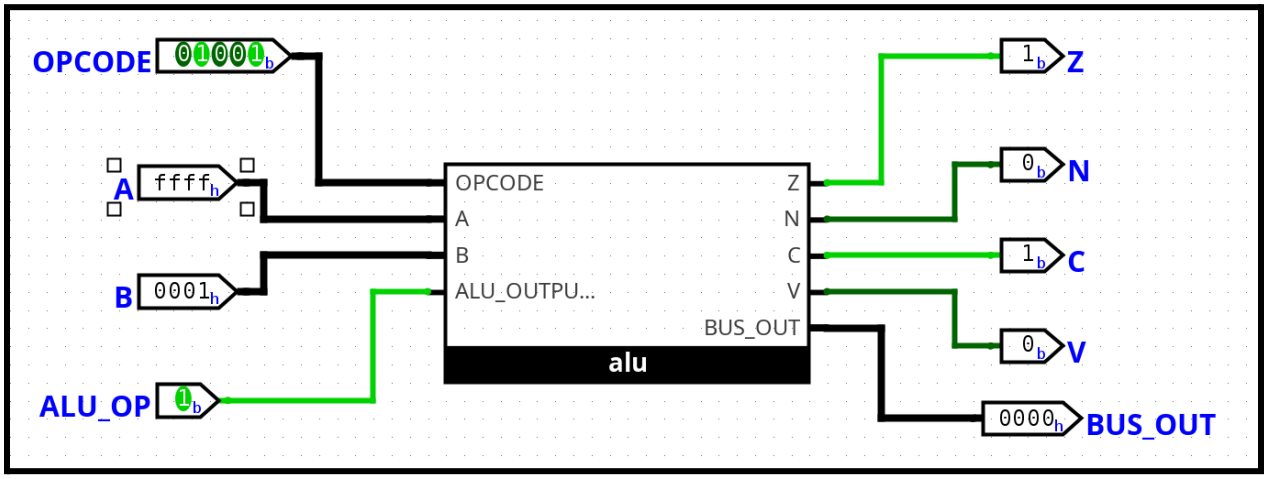
Design Decisions:
Using only 2to1 multiplexers, made a tree to use opcodes
Since sometimes, the ALU won't be required, there is an ON-OFF switch for the ALU.
Using comparator circuit done in the previous take-home assignment, checked if the output is zero, negative, has carry or overflow
Using a comparator buffer, I passed the resulting output.

Test evidence:

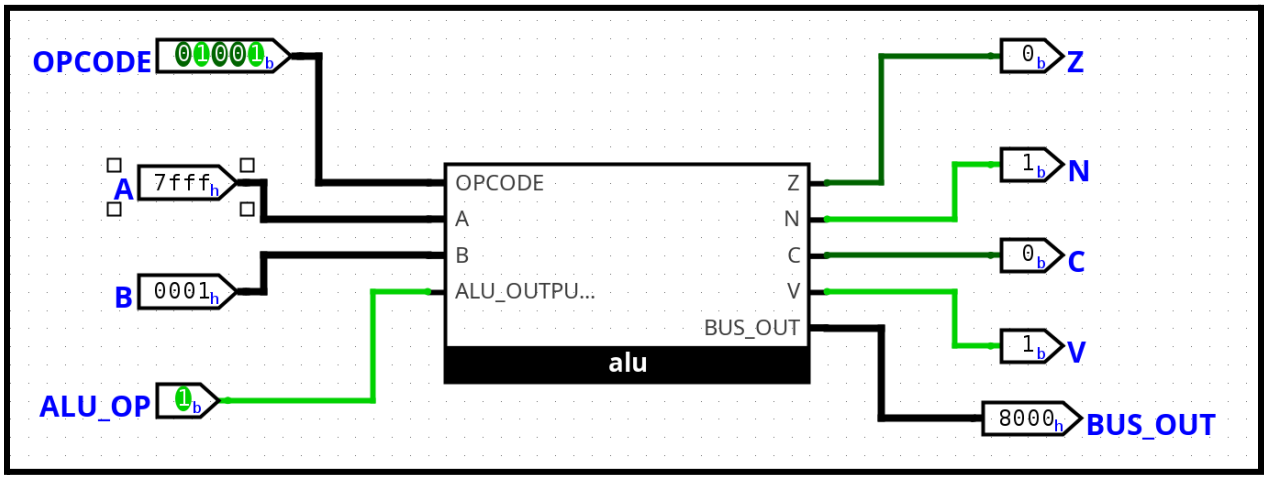
SUM



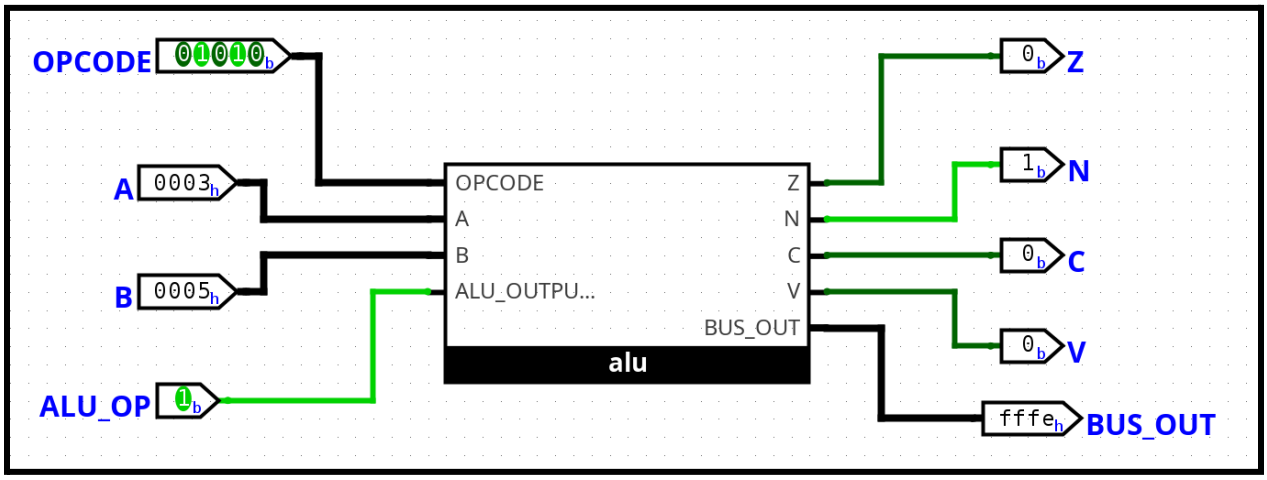
SUM with Z = 1, C = 1



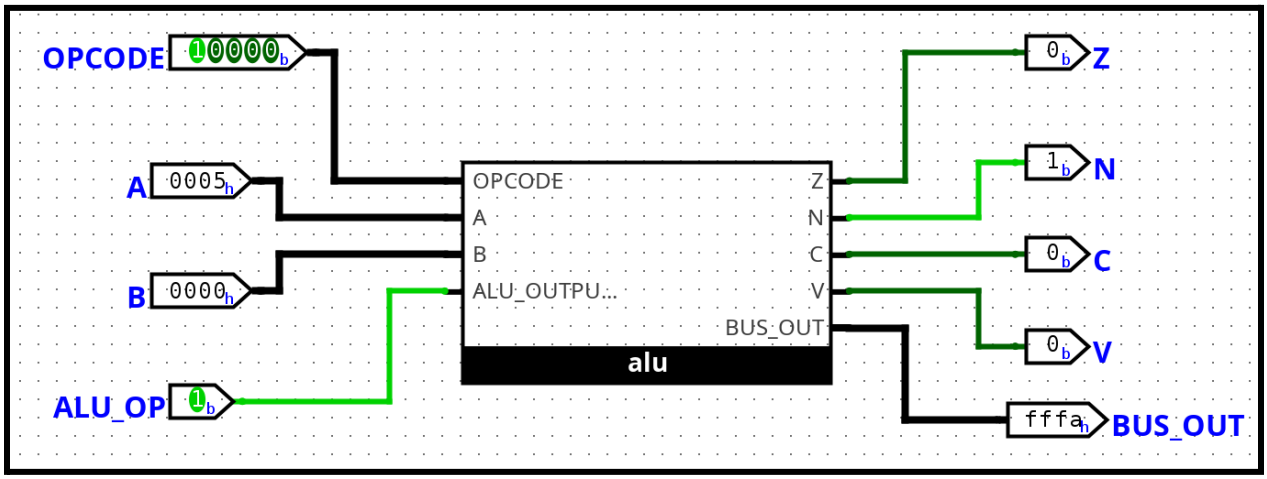
SUM with N = 1, V = 1



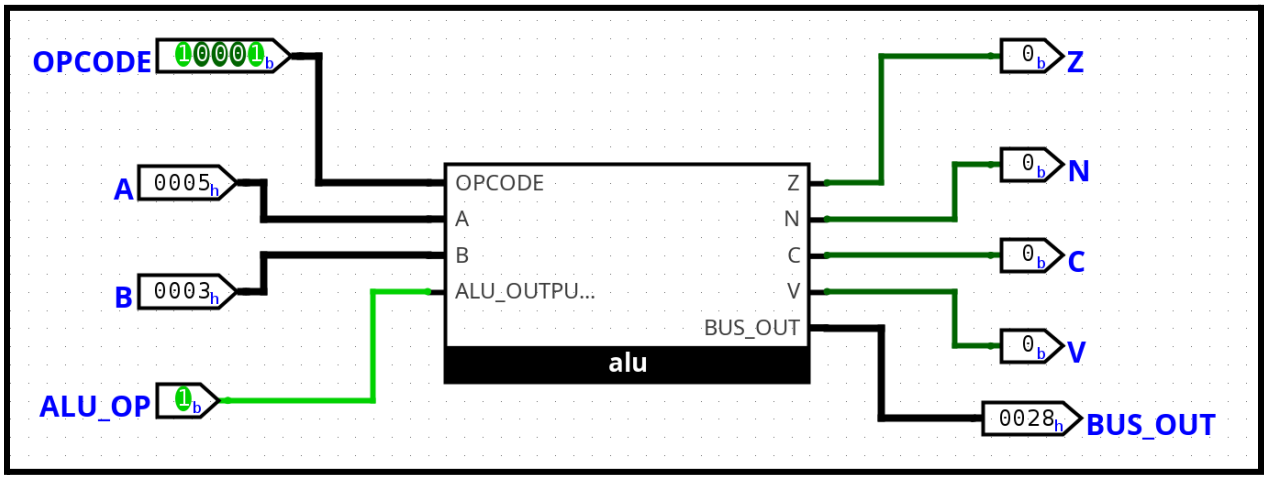
SUB



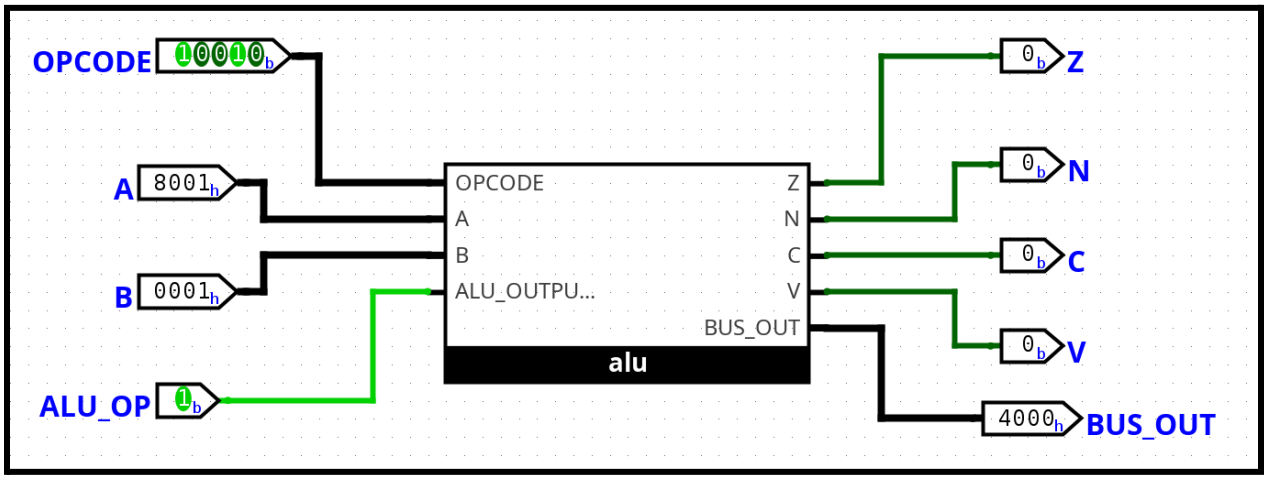
NOT operation



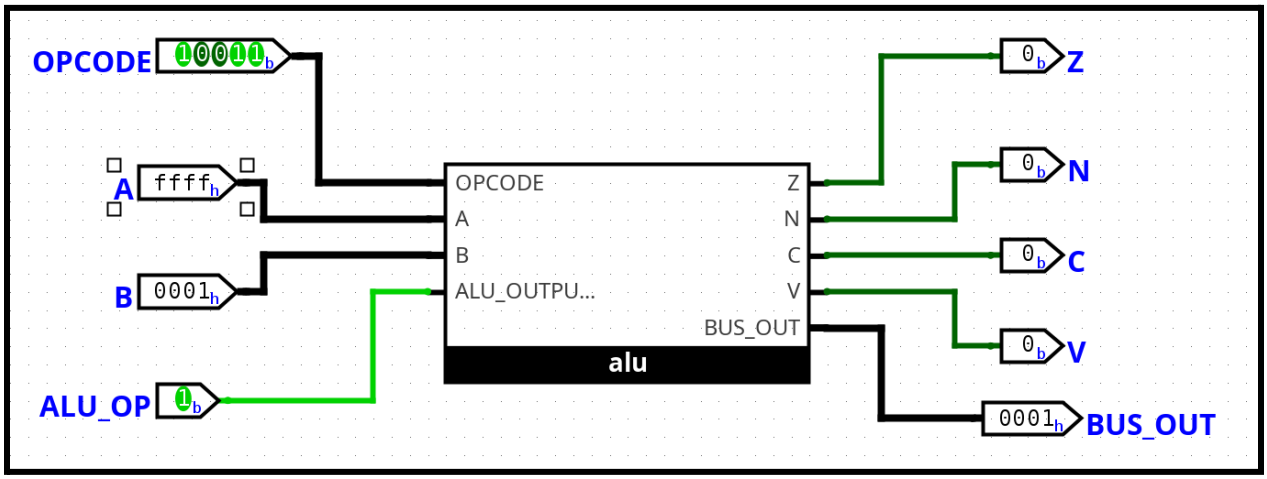
Left Shift



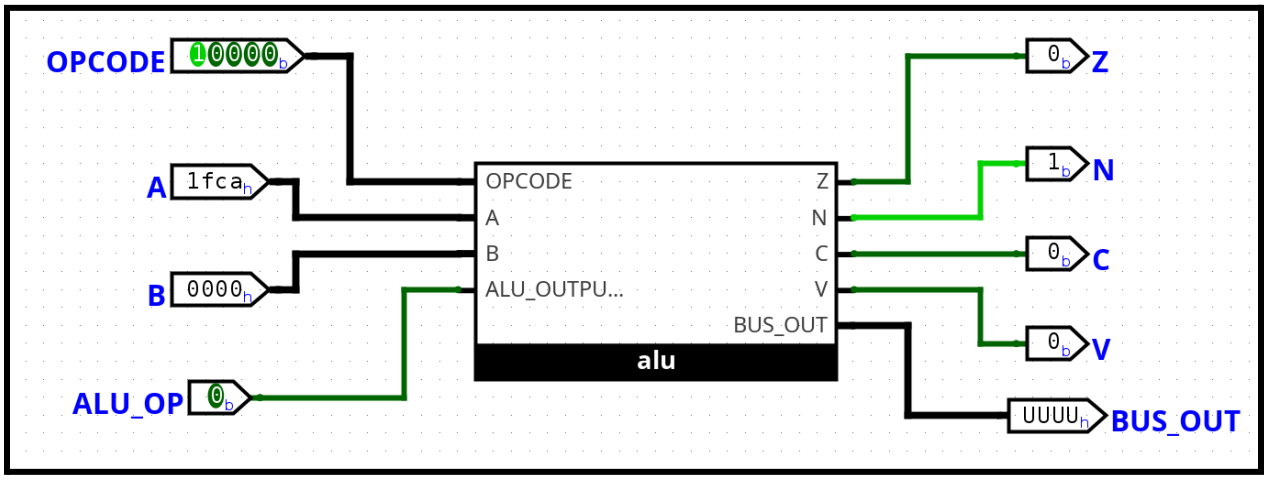
Right Shift



SIGNED LESS THAN



ALU = 0, High Impedance



Faults which occured:

There was one mistake of putting a NOT gate in front of COUT for SUB opcode which we had to fix.

Changed the fan-out of specific splitters and set the bits accordingly to make the circuit look cleaner